ripwire

The ripgrep of AI context.

Paddle out with a map.

See the rip before you're in it.

A zero-runtime-dependency C++23 CLI that maps any codebase into a ranked, deterministic call graph for coding agents — and puts a tripwire on every claim it emits.

rip — the speed half

Structural answers in tens of milliseconds, from a call graph it built itself.

wire — the other half: honesty

Unprovable totals ship labelled as floors. A zero means none found — never none exists.

Every truncation is disclosed, and every guess says how many it chose from.

Apache-2.0 · single binary · hermetic build (proven with the network off) · 27 vendored tree-sitter grammars

// the problem

Your agent reads the whole library to answer one question

Blind grep, then whole-file reads. Most of what enters the context window is never used — it is paid for anyway, on every step of every task.

And bigger context is not better context: irrelevant text actively degrades the answer. The job is selection — a small, ranked, structural map beats a pile of open files.

0.108 s

median answer, warm index — round 4 (August 2026, historical), all arms one machine one day, N = 60

−39.4%

token ceiling vs the un-routed baseline — LocBench cost ledger (August 2026), N = 243

Speed caveat travels with the number: ripwire answers from a warm, pre-built index, and round 4 measured its index cost too — 0.31 s, tabulated beside the competitors' rather than omitted. Measured 2026-08-08, before 0.6.0's performance work; not re-measured since.

// one binary, the whole pipeline

Crawl → parse → resolve → rank → emit. Deterministic, end to end.

crawl	→	parse	→	resolve	→	rank	→	emit
the tree, no VCS needed		tree-sitter, 27 vendored grammars		references → call graph		Personalized PageRank		minified XML, one line

byte-identical	two runs, same bytes — a gate on every push, not a tendency; warm equals cold
zero runtime deps	CMake + a C++23 compiler; builds with the network off — vendored everything
the languages	Rust · C++ · ObjC/C++ · C · Metal · CUDA · Python · Go · Swift · TypeScript · JavaScript · Java · Kotlin · Ruby · PHP · Lua · Elixir · Dart · GDScript · Clojure · Bash · C# · Protocol Buffers · JSON · TOML · YAML · Markdown — 27 vendored grammars, Astro on the TypeScript one; markdown and reST headings are real symbols
agent-native	an MCP server and 194 long flags behind one `--help` that is always the authority

// every feature, one screen

Seven families of questions it answers

understand cold	*What is this repo, and what matters in it?*	`--for` `--tree` `--lego` `--exemplar` `--recall` `--pack-task` `--token-budget`
navigate	*Who calls this? Safe to change or delete? Which tests?*	`--callers` `--callees` `--uses` `--defs` `--impact` `--path` `--connect` `--affected` `--situ` `--test-gate` `--from-trace` `--pattern` `--safe-delete`
detail ladder	*Show me more — but only where it pays.*	`--detail` `--pack-signatures` `--outline` `--expand` `--compress`
quality & risk	*Where is the risk, and did I just add some?*	`--quality-panel` `--quality-delta` `--dmm` `--biggest-first` `--ensemble` `--context-ratio` `--nonlocal-state` `--field-affinity` `--hotspots` `--lint` `--clones`
self-diagnosis	*Is my setup actually working?*	`--doctor` `--skipped`
security	*Is this agent skill file safe to install?*	`--scan-skill` `--scan-skills`
knobs & modes	*Shape, format, cache, budget.*	`--json` `--format` `--mcp` `--lsp` `--max-memory`

`--help` is generated from the binary's own flag table — 194 long flags; docs/COMMANDS.md carries an entry for every one of them, 173 with a recorded invocation and its output

// reach for it at the moment, not the manual

The reflexes: which verb fires when

Landing cold on a task

`--for="<task>"` · `--pack-task`

ranked, budgeted context in one call

Stack trace in hand

`--from-trace`

frames mapped to symbols, innermost body included

“Is it safe to change X?”

`--impact` · `--uses`

the whole blast radius, not just 1-hop callers

Just edited a symbol

`--edit-check`

did the contract change, who breaks — ~ms, warm

About to write a helper

`--exemplar` · `--grep`

imitate the house pattern; catch the duplicate early

Landing several branches

`--merge-scout`

pairwise conflict sites + a landing order

Before you call it done

`--quality-delta` · `--test-gate`

only what you made worse; exactly which tests to run

Handing the area on

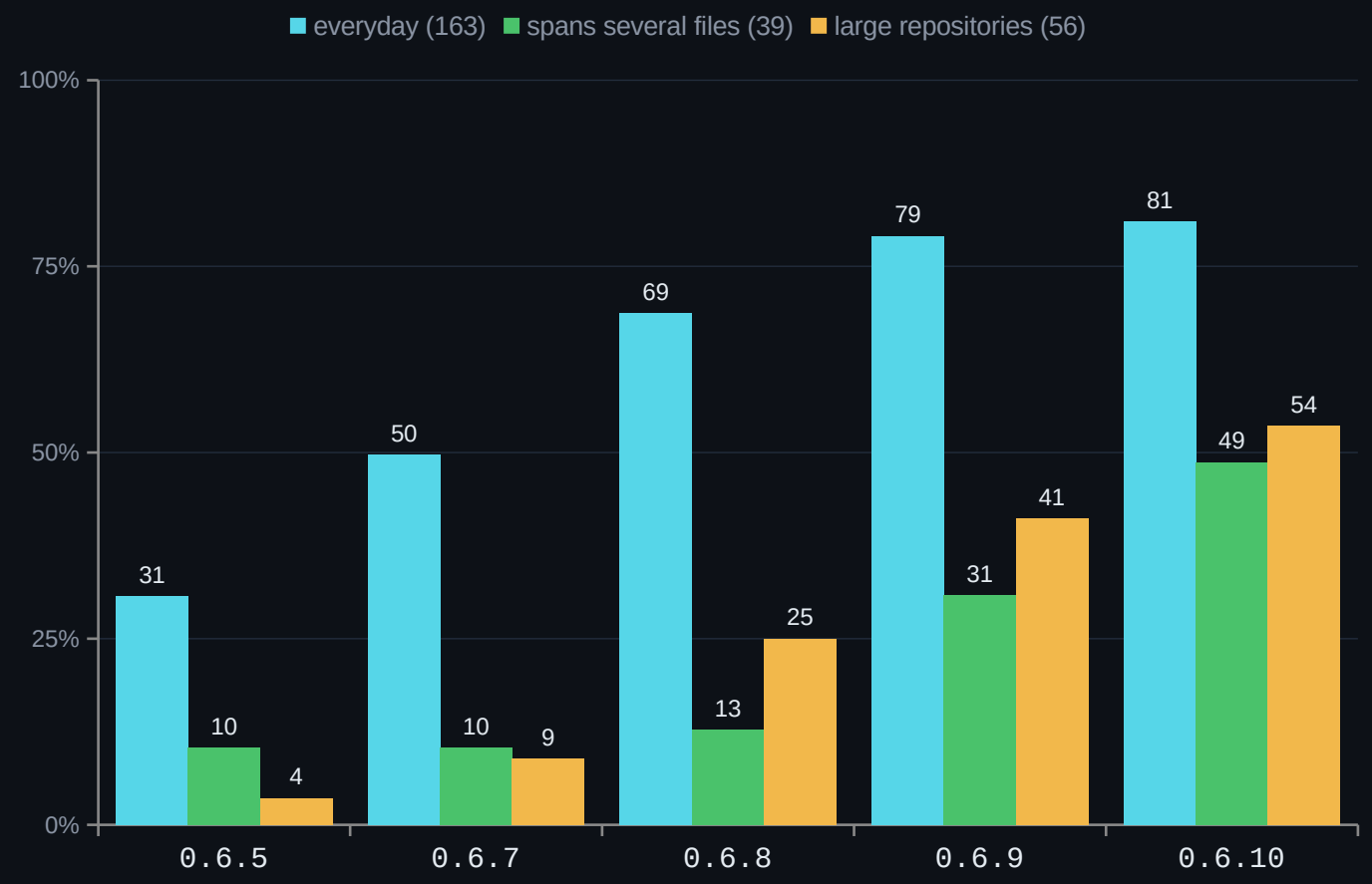
`--handoff` · `--note-add`

the verified-vs-heuristic packet the next agent needs

and `--pr-context` when you are reviewing someone else's diff — per-file blast radius, tests-to-run, hotspot flags

// same questions, commands and grader each release — in-sample

More answers complete in one call, release by release



% of each set answered completely: every required item in the first answer, nothing wrong

50 → 132

of 163 everyday questions complete, 0.6.5 to 0.6.10

3 → 0

everyday answers that stated something wrong; 0 in every release since 0.6.7

22.5 → 18.0 KB

bytes per complete everyday answer; it fell to 15.1 KB in 0.6.9, then 0.6.10's larger answers bought more required items

Read it as progress, not as proof. ripwire was developed against these questions. Large repositories: envoy, nushell, terraform, cpython, vscode, cockroach and zed. A held-out comparison, with other tools, comes with 0.7.0.

README.md, "Measured": the answers-by-release table; its footnote defines complete / partial / none

// 0.6.9 (2026-10-29) and 0.6.10 (2026-11-08); each row names its source

0.6.9 and 0.6.10: hedged, not wrong — and much lighter

hedged instead of confidently wrong

implementors	a call through an interface-, trait- or abstract-typed receiver names the implementations that can run, with a floor when the list cannot be complete (0.6.9)
no false coupling	--arch, --deps, --cochange, --impact and --export=cc.json stop reporting coupling from edges the index never had (0.6.10)
no namesake	a Rust prelude call, a macro with no in-tree binding, a Python builtin no longer bind a same-named definition (0.6.10)

answers that carry more

--defs	which definitions a name binds, with the spelling that picks each one alone; member rows print their owner (0.6.9)
--path bodies	the middle hop of a short path, the route an unreached path names; hops print call site and signature (0.6.9, 0.6.10)
--for reach	registrations and what is registered under a key, a literal's sites, a --flag and its default, an add-a-new-X family, re-export hops (0.6.10)

lighter on large trees (percent, per PR)

−35 to −67 %	warm CPU: vscode --deps −67 %, cockroach --for −57 %, envoy --deps −35 % (#542); TypeScript --deps a further −22 % (#554)
about −48 %	peak footprint of an index on macOS: envoy −49 %, cockroach −47 %, vscode −48 % (#497)
−30 to −40 %	a warm run's peak, now at or under the cold run's (#556); the include closure in an eighth of the memory (#505)

languages and reach

Rust · Go · C	functions stored in struct fields join the calls through them; Rust crate::, super:: and mod paths resolve in their own crate
C / C++	quote includes resolve at the include roots a build passes; pointers to member functions; using and namespace aliases
reST · Sphinx	documents become real sections; --recall serves the best-matching part of an HTML page (#409)

performance rows: each PR's own measurement vs the build before it, one machine, not measured together · caches rebuild once

// on main since v0.6.10 — changelog.d/, each with its own measurement

Since 0.6.10: less memory, cheaper cold starts

memory

264 → 144 B	per indexed reference: five fields empty on most rows moved to a side block. Peak −10 to −24 % on envoy, cockroach and vscode; answers byte-identical
the sum	this and every memory step since 0.6.9, measured release against release on the same three repositories: the next slide

speed

−7 to −15 %	cold --deps instructions: one cached root path per file instead of a re-descent per parent step (C++ −15 %, Rust −13.5 %, Go and Python −7 %)
−24 %	cold --deps instructions on vscode: the JS/TS side captures walk once with one cursor
−15 %	warm --for instructions on cockroach: clone keys computed at parse time, so a warm pass opens no file

a zero that means "not indexed" says so

--uses	on a struct or table in a language whose reads are not captured: unindexed_roles= instead of a bare count="0"
--exemplar	an off-task pick carries off_task="1"; --help-task abstains with a next step instead of nothing
--connect	never names a test function as the joint of two source symbols

new, and from the community

Protocol Buffers	messages, enums, services and rpcs are definitions; a generated name links to its schema by name
--deadline-ms	prompt hooks no longer time out on every prompt in a cold directory: reported with measurements by @WingedGuardian (#594)
Py · Rust · C++	from pkg import mod draws its file edge (@jflowers, #533); a Rust re-export (@soarkit, #610); C++ bases as captures (@SeddonShen, #538)

caches rebuild once on main (parser 307, cache 50) · the fragments fold into the next release's CHANGELOG entry

// warm --deps peak memory by release — percent change, one machine

Warm peak memory: about three quarters lower since 0.6.9

peak footprint, % of v0.6.9's (shorter is lighter)



Why it is a headline

Every megabyte the tool holds is a megabyte not available to the agents running beside it. On a large repository, memory decides whether a code-graph tool can run on a laptop at all.

How it was measured

Dev builds of the two release tags and main; one shared 16 GB Apple M2 Pro laptop; peak memory footprint (macOS /usr/bin/time -l), median of three warm runs per cell, binaries interleaved. Percent only: absolute megabytes come with 0.7.0, measured on another machine.

docs/EVALS.md, "Memory by release": method, table and caveats · per-PR steps: CHANGELOG [0.6.10], changelog.d/

// the write-time check: what YOUR change made worse — and only that

Quality delta: twelve kinds, and a gate that only fires on old code

complexity	verbosity	nesting	params	duplication	dead-code
api-surface	error-masking	short-horizon-churn	new-clone-of-reused-helper	placeholder	defect-shape

exit 2	only when code that EXISTED at the baseline got materially worse, and nobody acked it. That is the gate — a regression you introduced into old code.
new code	rows for a symbol that is new are printed, never gated: they are the debt you are adding, to read. Exit 0 means nothing that already existed got worse, not "clean".
one exception	defect-shape format-arity gates on any origin: a format string whose arguments do not match is a bug, not a taste. Measured: 0 false positives over 1,330 literal-format calls on this tree and 405 in the Python 3.13 standard library.
never gates	placeholder — the stubs and TODOs a change adds — is an honesty check on "done": listed per symbol, never a failure.

Why only old code gates: the change an agent must not make is the one that quietly degrades what already worked. Its new code is shown to it at "done".

docs/EVALS.md §6, "--quality-delta's twelve kinds" (format-arity measured 2026-10-07) · ripwire --help, --quality-delta

// 0.6.0 — everything since v0.5.0 (tagged 2026-09-07); each figure names its PR

0.6.0: faster where it hurt, clearer where it stops

two new languages

Dart	the 23rd grammar, by @calvinchengx: 71,726 symbols from flutter/packages' 3,706 .dart files, none among its 289 degraded parses (#75, #106)
Kotlin	by @xCatG: calls resolve between Kotlin and Java, both ways; no sanitizer finding across 501 real .kt files, deterministic on ~9,500 more (#126)
any language	Dart's landing caught six per-language arrays sized off the last enum by hand: an appended language dropped out of --skipped's census silently. Fixed for all.

Rip'n Fast

159.7 → 9.2 s	warm --grep on llvm-project: an inheritance cone rebuilt on every still-ambiguous receiver-typed call is now built once per type (#83)
−19.9% CPU	cold parse on llvm-project, 194.1 → 155.6 s: child walks that went quadratic now use a cursor, output byte-identical (#127, #130)
−27.6% CPU	warm --pack-task on go, 8.13 → 5.88 s: the tokenizer rebuilt on one header of NEON, AVX2 and scalar string kernels (#127)

honest where it counts

2,107 → 3	false callers of memgraph's SafeString::move: a std::move call no longer binds to your own code (#134)
12/12 → 8/12	commits --quality-delta gated, of 12 landed: true-positive share 2% → 12%, with 0 wrong rows (#127)
13/25 → 0	harmful --help-task recommendations on the adversarial prose set; precision 0.797 → 1.000 (#127)

fewer tokens, nothing hidden

46,385 → 4,473	tokens for --help: one line per flag now, and every disclosure is still one call away (#92)
6 → 50	--handoff symbols per code file (12 per prose file): containment 16% → 54%, purely additive (#127)
pages, not cuts	--doc-drift, --flags, --flip and --situ disclose their cuts and page; --situ lists every tests-to-run row (#127)

upgrade note: prebuilt x86-64 binaries now need an x86-64-v3 (AVX2-class) CPU, RHEL 10's own floor — the installer checks before it downloads and names what is missing; arm64 uses NEON (#127, #137, #138)

// 0.6.1 — everything since v0.6.0 (tagged 2026-09-07); each figure names its PR

0.6.1: it resolves what it names, and says what it cannot prove

languages that resolve, not just parse

module · name · arity

Elixir calls resolve to the module, name and arity they name, not by name alone: lexical aliases, filtered imports, default arguments, pipes, captures, delegates. Macro expansion and unquote stay documented limits (@henry-hz, #207)

none → 79%

of occurrences matched on spring-petclinic: SCIP encodes a range two ways and ripwire read only the deprecated one, so every scip-java index was silently ignored and --scip changed nothing (@dpunosevac, #198)

it says what it could not prove

89 of 4,322

answers shrank over every file:name selector on ripwire's own tree that selects only declarations — and NONE grew. A header selector kept a definition only when its file is the header or includes it, resolved path-precisely (#173)

nine listing verbs

now count what the proof dropped as unproven_defs=; all but two had answered from the declaration alone and printed a clean zero (#190, #195)

not a safe edit

an incompatible="0" beside unproven_defs= is now STATED to be an incomplete read, on every verb that resolves a focus symbol, --edit-check included (#210)

every number carries its definition

graph_unindexed=

shipped in 0.6.0 with no definition on three verbs, and under a compact legend on every XML verb but one (#169)

and eight more

declined_calls=, unproven_defs=, pr_iters=, the map header's counts, --impact's blast-radius counts, --safe-delete's verdict fields and the community structure counts: each defined now (#185, #189, #203)

pins follow prose

the compact byte pins now follow the definitions, instead of a definition being trimmed to fit one pin; a budgeted --for that drops legend clauses NAMES the attributes it dropped (#174)

it fits in memory at llvm scale

114 MB → 368 KB

the declined-call index on llvm-project. Candidate lists were stored once per call, so the structure grew with calls x candidates: 27.9 M entries. One copy per DISTINCT list makes it 9,879 lists and 62,359 entries (#208)

every byte the same

counts and output are unchanged by that change — the win is the structure, not the answer (#208)

the release ran against instruments registered before the work began: a frozen bank of 30 retrieval questions answered in ONE call on a 2,066-file corpus pinned at one commit, a six-step follow-up ladder with no model in the loop, and a separately registered held-out draw so a round cannot be tuned onto the bank it is graded on

// 0.6.2–0.6.5, 2026-09-14 to 09-30; each row names its PR

0.6.2 to 0.6.5: more places it runs, more code it answers for

where and how it runs

Windows x64	builds under clang-cl and MSVC cl.exe; a release zip ships as a preview. The gate suite does not run there yet (@lennix1337, #44)
--lsp	a read-only navigation LSP server over stdio: definition, references, document and workspace symbols, hover (@mpapis)
65% · exit 5	a zero-config memory guard: past 65% of RAM or the cgroup limit, a disclosed partial map or exit 5, not a swap storm; --max-memory moves the line (#350)

better answers

63 → 2,083	functions --quality-panel measures on a 583-file TypeScript repo: a name bound to an arrow or function literal now owns its body (JS, TS, Lua, Python)
one row per def	--metrics: overloads and #ifdef twins each get their own row and metrics, instead of sharing one body's
Ruby, from the community	class Child < Parent is an edge, so --lego answers for Ruby (#325); RSpec's described_class pins to the class under test (#338), @andriytyurnikov
72.8% of vue-core	call sites had no caller, measured before the fix: a call outside every named function now has a module-scope owner (#60)

honest answers

rank, then cut	capped verbs keep their strongest rows and name what they dropped (0.6.3); three answers that cut silently now say what they left out
shallow="1"	on a depth-1 clone, the CI checkout default, the history verbs say the history is shallow instead of calling the boundary commit a root
five false answers	--flags, --doc-drift, --from-trace, --connect and --dead-code each answered falsely on a real repository; each is fixed

quality you can act on

placeholder	the 11th --quality-delta kind: the stubs and TODOs a change adds, per symbol. It lists them and never gates
d= hop depth	--impact lists the blast radius nearest first, so a cut page drops the deepest rows first; by_depth= says where it stopped
--mcp-tools	the MCP server lists only the tools you name, or a core profile of 8; the full 33 stay the default, byte for byte

upgrade note: parser 129 · cache 27 — a cached index from an earlier build re-indexes once on first use

// Rip'n Fast, at scale — llvm-project, **182,555** files, as the instrument

Three bugs found at llvm-project scale

O(C²) child walks

An indexed ts_node_child(n, i) loop re-walks the child list on every call, so wide, flat lists (comment runs, C/C++ include guards) go quadratic. #127 moved 24 walks to one cursor helper (18 arms red first at 11–125×); #130 moved 22 more (13–126×).

194.1 → 155.6 s

cold parse CPU, n = 1
−19.9% · wall −25%

the inheritance cone

Rebuilt on every still-ambiguous receiver-typed call: 86,667 cones for 2,984 distinct types, 143 s of a 154 s run. Each type's cone is now built once, and default maps are byte-identical on go and llvm-project.

159.7 → 9.2 s

warm --grep
default map 248 s → 10 s

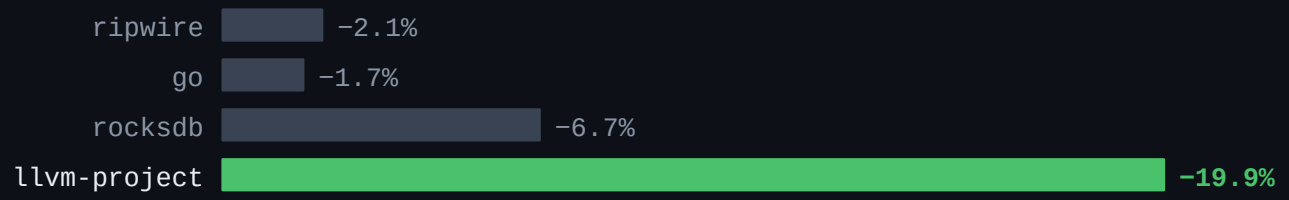
the cache that evicted itself

One llvm root needs 1.76 GB of cache, and the 2 GB oldest-first sweep was evicting that root's own index. The sweep now pins the root you are working in, and two key builders that hashed one root with different seeds share one key.

274 → 26 s

a repeated --for, CPU

the same round's cold parse, CPU Δ by corpus (median of 5; llvm-project n = 1)



Why llvm-project

A scale test that is also the real thing. Built on or with LLVM:

Clang (C, C++, Objective-C)
Chromium's only supported compiler; the Android NDK's; Apple Clang in Xcode; the PS4 toolchain

Swift · Rust · Julia · Zig · Kotlin/Native
each has an LLVM backend for its code generation

CUDA
NVIDIA's NVVM compiler is based on LLVM, and Clang compiles CUDA too

Metal
Apple's shading language: “Metal uses clang and LLVM”

MLIR
part of LLVM; used by TensorFlow, OpenAI Triton and Mojo

WebAssembly
Emscripten, a compiler toolchain to WebAssembly using LLVM

ROCm · oneAPI
AMD ROCm's compilers fork llvm-project; Intel's oneAPI DPC++/C++ uses LLVM

LLVM received the 2012 ACM Software System Award · each item's source is in the notes

all three measured on llvm-project; no standard corpus could see the first · sources: #127, #130, #83 — quotes in the notes

// wire, in practice — three stories from 0.6.0, all merged

Confidently wrong, caught, and fixed in the open

#134 · the resolver

std::move had 2,107 callers

On memgraph, calls written std::move bound to the repository's lone in-repo move, SafeString::move, at full confidence and with no amb= or prov= marker. It became map row #1. In ripwire's own graph, std::min, std::max and std::sort bound fastmath and svector members.

2,107 → 3

callers of SafeString::move after the fix: two of its unit tests and one std::ranges::move, a disclosed gap

what catches it now

A std::-qualified call keeps only candidates inside namespace std; anything else counts toward external= and gets no edge. On ripwire's own tree, --callers=min went from 131 to 5.

test/stdqualcheck.sh · 35 checks, 19 red before

#127 · the string kernels' gate

The probe that lied

strkerncheck asked Rosetta 2 whether it could run the x86-64-v3 kernels, using a one-add AVX2 probe. Clang folded that probe to a scalar addb, with zero ymm or VEX opcodes, so it answered ok on every runtime while the v3 slice SIGILLed on the macos-14 runner. The first fix's commit gave the wrong reason; the next one corrected it.

0 ymm opcodes

in the probe binary that had answered “ok” (otool)

what catches it now

The probe is now built with the floor's own -march and touches every v3 extension. The gate disassembles it and requires ymm, pdep, pext, lzcnt, tzcnt, vfmadd and vcvtph2ps; a folded probe routes to the scalar slice instead of passing.

test/strkerncheck.sh · b4ebf0bb, 150fb6d3

#127 · the audit's own tools

The audit's instrument was wrong twice

A 14× super-linear reading turned out to be the cache evicting its own working root. And the recorded cap sweep had measured retrieval caps on a population of zero: it discarded exit codes, counted rows that emit nothing, and read back its own 10.4 MB cache blob, which “responded” to 103 of 108 caps.

59/195 → 64/151

cap-sensitive rows: the published split, then the split re-derived over the rows that answer

what catches it now

Every sweep row now carries a state (ok, rc=N, unparseable, unexpanded, timeout), a run where nothing answered refuses to report, and nothing it writes may land in the corpus. The cache sweep pins the root you are working in.

test/capsweepcheck.sh · 209f97a4 · 5723b2c0

every figure is quoted from its PR body or commit in the speaker notes — the tripwire covers our own instruments too

// wire, in practice — three more: the parser, the resolver, and a new language

Three more: a semicolon, a silence, a new language

#135 · the parser

One missing semicolon

A member macro with no ';' as a class's last line, like EXC_NAME(Foo) before '};', derails tree-sitter's C, C++ and ObjC error recovery. On memgraph one file had 472 of its 487 definitions misfiled, and a 14-line function ranked #4 in --hotspots, with nothing on the row saying so.

ccx 920 → 2

PrintFuncSignature, 14 lines: cx=749 ccx=920 before, cx=3 ccx=2 after the re-parse

what catches it now

Misfiled definitions carry extent_suspect=, and --hotspots leaves them out and says so. An error-gated re-parse blanks ALL-CAPS member macros and keeps the result only if it has fewer error bytes. Lowercase and namespace-scope runs still derail: flagged, not fixed.

extentcheck.sh (55) · macroreparsecheck.sh (103)

#136 · the resolver

22% of calls, declined in silence

--callers answered count=0, as if nothing called. When a call names two or more same-named definitions, none in the caller's file or directory, the resolver declines to guess: the right rule, but the decline was silent. On memgraph, 22% of all call references were declined this way.

65,516 / 295,086

memgraph call references declined (22.2%), now shown as declined= and declined_calls=

what catches it now

Every call reference ends in exactly one named bucket, and --pin-census prints the balance. It proved itself in the merge: #134's std:: refusals had gone uncounted, 1,495 on ripwire and 4,966 on memgraph. Now they count, and unaccounted=0.

test/declinecheck.sh · 17 languages, 104 checks

#126 · Kotlin, by @xCatG

A new language must not change old answers

Kotlin brings calls that cross into Java. In #126's review, once retrofit's same-named Kotlin body() methods were indexed, calls to the Java Response.body were declined, and its callers fell from 279 to 5: a Java-only answer, changed by files in another language.

279 → 5 → 279

callers of retrofit's Response.java:body: before Kotlin, with the first Kotlin binary, after the fix

what catches it now

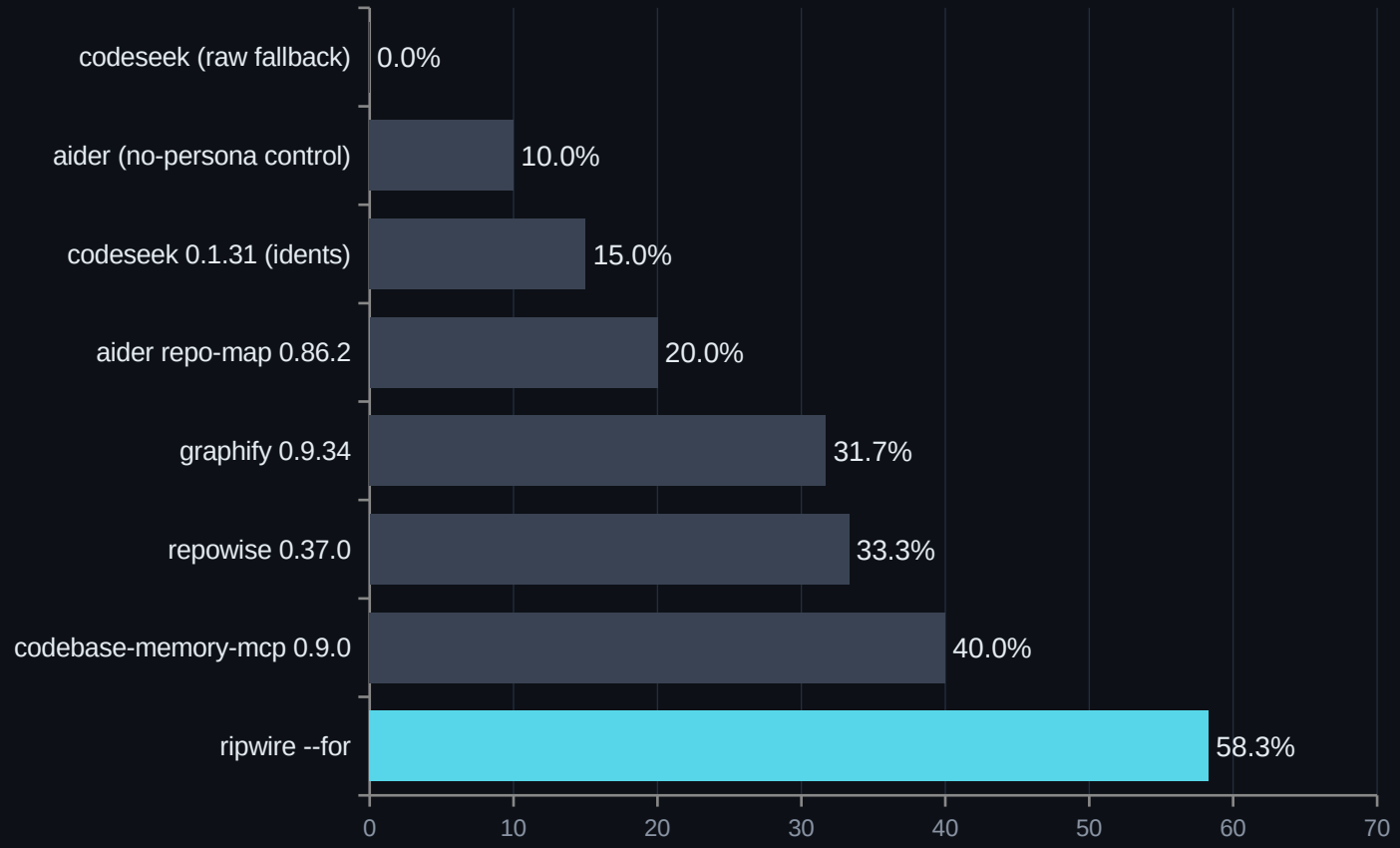
A call reaches the other JVM language only when its own language has no candidate, so adding .kt files never changes a Java-only edge. The bridge keeps its real job: a call into a name only the other language defines.

test/kotlincheck.sh §14 · the Java-only invariant

every figure is quoted from its PR body or commit in the speaker notes

// one harness, one machine, one day — N = 60 paired, zero exclusions

Head-to-head: every competitor, one table



1.46×

the margin over the best competitor — 58.3% against 40.0% strict file@10

And the index it was measured with
ripwire 0.31 s · codebase-memory-mcp 1.24 s · codeseek 3.37 s · graphify 7.82 s · repowise 34.0 s, whose worst single index was **352 s**. Cold, nothing to an answer: 0.213 s.

What re-running cost us, published: aider moved 18.3% → 20.0% on its own tie-break nondeterminism, and we printed the higher number. Paired, ripwire loses 2 instances to codebase-memory-mcp, 2 to repowise, and 1 each to graphify, aider and the aider control.

bench/headtohead/r4-2026-08-06/ — harness, JSONL and recipe committed; replaces the two non-comparable tables r1 and r2 needed · measured 2026-08-08, BEFORE 0.6.0's speedups: timings not re-measured

// scored against a third-party oracle, never against each other

Graded by a compiler: zero silent misses

Both tools were scored against a scip-clang compiler-grade index of this repository — 68 answers, 34 blind-authored queries × --uses and --callers.

0

true silent misses — pre-fix and post-fix alike. Every imperfect answer either carried a discriminating self-flag (amb=, defs>1, external="1") or was an oracle-scope artifact where ripwire was right and the compiler could not see the file.

What a compiler-grade index costs

	ripwire	Serena 1.6.2.dev0 (clangd 19.1.2)	
warm query	**49.8 ms**	3,760 ms	~70×
cold start	**208 ms**	8.7 s	~40×
index build	**none**	~8 min	—
errors	**0 / 101**	7 / 61	—

Read honestly in both directions. The LSP is more precise — site-level 0.929/0.941 against ripwire's 0.914/0.941 after the fix round, a gap of roughly 1.5 points with recall now equal. ripwire also pays ~7× the bytes per call, because its output carries self-describing legends. What it cannot do is answer a macro query at all — clangd's documentSymbol has no macro entries, which is 3 of its 7 errors — or start in less than eight minutes.

The number that moved the wrong way, published with the rest: over-hedging rose 18.6% → 22.6% across the fix round. Closing loss buckets added self-flags faster than it removed imperfect answers. That direction is deliberate — calibrated to warn too often rather than too rarely — but it is a cost, and it is on the slide.

bench/headtohead/r9-2026-08-09/RESULTS.md — C++ only, 34 scorable queries, one corpus (this repository)

// 48 paired questions, zero exclusions, both arms exit 0 on all 48

Against a graph-database context server: 27–7–14

class	n	ripwire	GitNexus	tie	bytes ÷ it
symbol lookup	12	**6**	2	4	**1.35×**
conceptual search	15	**8**	1	6	**1.23×**
callers / blast radius	12	**7**	2	3	**0.39×**
task orientation	9	**6**	2	1	**0.06×**
all 48	48	**27**	**7**	14	**0.159×**

Its seven wins are named one by one, and its first-pass numbers do not exist. Four are cost losses on an answer ripwire gets right — a by-name lookup it serves in ~1 KB where ripwire spends 3× to also hand back the body. Three are ranking misses on webpack, and two of those share one unfixed mechanism: a symbol whose NAME matches the query beats the implementation that carries it in its body. Under the improve-first rule the round's FIRST pass was never published — its product was a loss list, and what is on this slide is the state after that list was worked. One of those fixes met its pre-registered band and was reverted anyway for failing a separate standing requirement; **the loss it would have fixed is still counted against us above.**

The cost of the whole sweep

309,348 B against 1,945,231 B. Warm median 197 ms against 1,082 ms. Index 0.25–0.45 s / 6.6–16.5 MB against 23–52 s / 391–623 MB — written into the repository, not $TMPDIR.

What it does better: compact one-hop context at 0.7–1.0 KB, a depth-LABELLED blast radius, per-stage timing on every query, and two-command onboarding into eight agents.

Failure modes, both directions

needed a 2nd call	**0**	8/48
empty radius, real callers	**0**	1
malformed through a pipe	**0**	7/48

The published numbers use the file-redirected form — the configuration favourable to it.

docs/EVALS.md §2 — GitNexus 1.6.9 (npm latest at 2026-08-22) vs ripwire at 7eb638e; django, webpack and this repository, all tree-hash verified clean · ratios are on class TOTALS, the medians form is printed there too because the two disagree

// the rule: numbers are held back until the loss list is worked

Every head-to-head round, and the ones that went against us

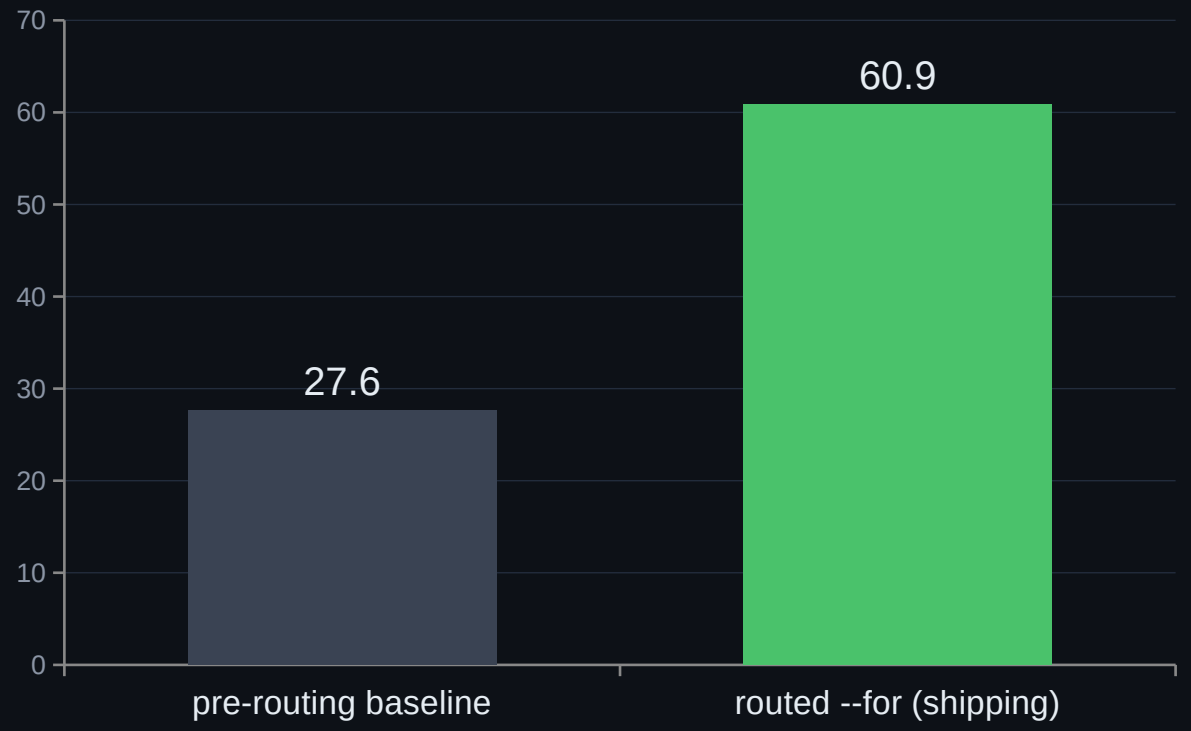
r1	2026-07-13/14	Aider repo-map · codebase-memory-mcp · graphify	*superseded by r4*
r2	2026-08-03	repowise · codeseek	*superseded by r4*
r3	2026-08-03	headroom — a compression layer, so a different instrument	passed code through byte-identical
r4	2026-08-06/08	all five competitors, one harness, one day	the table on the previous slide
r6	2026-08-04/05	agent-in-the-loop: the Codex CLI pilot	localization parity; +80% tokens, classified
r7	2026-08-08	CodeGraph, loss-first — we looked for our losses	fix REJECTED by its own preregistered band
r8	2026-08-08	aider again, at equal token budget	loss questions traced to one root cause
r9	2026-08-09	Serena / scip-clang oracle	0 silent misses; the fix list it produced
r10	2026-08-22	GitNexus 1.6.9 — a graph-database context server	27-7-14 — its own slide, two back

r7 in full, because it is the point: the fix was built, gated green, measured — predicted 12/22 against a pre-registered band of 10–14, measured 5/22 — and reverted rather than tuned. Published as a negative in docs/EVALS.md §7.

bench/headtohead/ (r1–r4, r9) · bench/r7/ · bench/r8/ — harnesses committed even for rounds that rejected their own fix · r5 left no head-to-head record in this tree, so it is not listed · r10's method and pins are published in docs/EVALS.md §2, its pre-fix numbers nowhere

// held-out, repo-disjoint, frozen dataset

Routing the ranker: +33 points, measured properly



+33.33pp — paired delta, 243 held-out instances in 78 repository clusters

+25.00pp — clustered-bootstrap 95% lower bound — the honest floor

−39.4% — token ceiling for that gain

+3.4% — warm latency for that gain

bench/locbench/ — czlll/Loc-Bench_V1, 560 rows, JSON SHA-256 pinned in dataset.lock; metric code = LocAgent's strict Acc@k

// compiled, not interpreted — measured on public corpora, August 2026

Fast enough to call on reflex

0.213 s

cold — nothing to an answer: parse, rank, reply, no cache (r4)

0.108 s

warm query, median — against aider's 2.920 s (r4)

0.31 s

index build — against repowise's 34.0 s, worst case 352 s (r4)

49.8 ms

warm query on this repo — against a clangd LSP's 3,760 ms (r9)

Where the second goes

Cold profiles are tree-sitter parse plus tag-query capture — not graph math. Resolve and PageRank together are a small fraction of the run, and the ranking was never the bottleneck.

Which is why the wins came from removing work, not micro-optimising it: demand-driven side captures, five whole-AST passes fused into one pre-order walk, a calibrated per-worker parse reserve.

Two opt-in faster builds

LTO is on by default. PGO is a script away and buys 14–25% cold.

Both are opt-in performance, not opt-in correctness: **the output stays byte-identical across build flavours** — the determinism gate runs on both, and CI builds a plain flavour too, because NDEBUG once compiled a whole class of degrade-path checks out of existence.

every card above comes from a committed harness — bench/headtohead/r4-2026-08-06/ and r9-2026-08-09/ (August 2026, before 0.6.0) — later speed and memory work is on the release slides, in percent

// docs/EVALS.md §5, pinned on this repository 2026-08-08 (the --for row re-pinned 2026-08-22)

Ten everyday moments, and what each one costs

Ask it	Command	ripwire	naive read	savings
Orient me in this repo	`ripwire .`	**~5.6K**	~20K–25K	**3.6×–4.5×**
Where is X handled?	`--for="…"`	**~2.1K**	~4.9K–20K	**2.3×–9.3×**
What do I already know?	`--recall="…"`	**~15K**	~445K	**29.2×**
Set me up for this task	`--pack-task="…"`	**~2.1K**	~16K–80K	**7.7×–37.7×**
Show me this one function	`--expand=SYM --top-k=0`	**~260–16.5K**	~43K–174K	**2.6×–670×**
Who calls this function?	`--callers=SYM`	**~580**	~40K–52K	**69.2×–89.1×**
Is it safe to change this?	`--impact=SYM + --uses=SYM`	**~1.3K**	~18K	**14.4×**
I have a stack trace	`--from-trace=FILE`	**~1.4K**	~124K–298K	**86.9×–208.6×**
I changed these files — tests? radius?	`--situ`	**~410**	~3K–132K	**7.3×–324.2×**
Review this PR/diff	`--pr-context=REF`	**~1.9K**	~4.8K–51K	**2.6×–27.5×**

Figures are ~tokens (≈ bytes/4). Every row is scored same-correct-answer-or-it-doesn't-count — both sides were checked, not assumed — and every ratio is a **range**: the cheap end and the honest end of what an agent would actually read, never the single most flattering number. Absolute counts belong to the corpus AT THE PIN — .gitignore became a crawl default on 2026-09-03, which moved what a re-run reads; the commands, not the constants, are what reproduces.

// the cost lever

Smaller answers that are also better answers

74.7%

fewer element bytes at top-50 — the --pack-signatures ladder, root-neutralised, re-derived on this repo every run

This figure survived its own audit: three corrections deep, re-derived root-neutrally after the original was shown to depend on how the corpus path was spelled — three spellings of one root read 18.6 points apart before that subtraction. top-50 is the quotable number because the payload is top-50 whatever --top-k says.

Ask for a symbol by NAME and the router uses the name-exact ranker:

```
MRR  0.745 → 0.960
recall@1  61.1% → 91.3%
```

name-shaped queries on src/, all 3,011 doc-commented symbols scored, re-pinned 2026-09-05

And the pollution metric — fixture and generated paths contaminating results — is driven to 0.0% on the ranking lane while recall went UP, not down.

docs/EVALS.md §4–§5 — showcasecapturecheck re-derives the byte-reduction triple; `ripwire src --eval-retrieval` re-derives the ranking bars, scoring every doc-commented symbol rather than a sample

// the half no other tool ships

The tripwire: honesty as an output contract

counts_floor="1"

call edges come from source text — dynamic dispatch adds no edge, so every count that cannot claim completeness is labelled a floor, in the output itself

a zero is a measurement

zero means none found, never none exists — and absent ≠ 0

truncation is disclosed

every cap, page seam and budget cut is named in the header before you read a row

refusals teach

a refused query names the flag, the problem, and a working example — never a bare error

```
$ ripwire . --callers=rankGraphTeleport

<callers of="rankGraphTeleport"
  defs="1" count="7"
counts_floor="1">
<s t="fn" n="getIndex" .../>
<s t="fn" n="rankGraph" .../>
…
</callers>
```

The map grades itself before it answers. This repository's own src/: files=275 symbols=14424 edges=24953 ambiguous=2290 unresolved=3211 declined=36734.

docs/EVALS.md §8 lists the numbers this project refuses to publish, each with its reason

// the honesty marks stop being a promise and become a measurement

Four levers accepted, one rejected on purpose

macro edges	`kParserVer 48`	function-like #define invocations land role="macro" edges on disclosed-degraded symbols
fn-pointer bindings	`kParserVer 49`	calls through unambiguous local function-pointer bindings now resolve
using-declarations	`kParserVer 51`	re-exports emit role="import" references — r9 loss bucket 1
shadow suppression	`kParserVer 52`	a local variable shadowing a function name stops stealing its use-sites — r9 loss bucket 2
RTA-lite	`reverted`	instantiation-filtered CHA cone: refuted TWICE — the instantiated set crossed namespaces, and narrowed calls LOST their amb= mark

The two r9 levers, against the compiler oracle: site precision 0.9046 → 0.9136, recall 0.9285 → 0.9412.

The fn-pointer lever raised unresolved= by 1,039. Disclosure, not regression: call sites the resolver now RECOGNIZES but refuses to resolve are counted instead of silently ignored.

`bench/resolverround/RESULTS.md — per-lever attribution; RTA-lite's rejection is written up in docs/EVALS.md §7`

// how it stays true

Proven, not promised

807 gate scripts

the suite runs on every push — plus determinism, cache-transparency and golden contracts; the gate count itself is a dated snapshot of the runner's own loop (counts snapshot 2026-11-13)

byte-identical, always

two runs over the same tree produce the same bytes; warm equals cold. Enforced in CI, twice — Release AND a plain flavour, because NDEBUG once blinded a whole class of checks

differential refactoring

a refactor must prove it changed nothing observable: two binaries, hundreds of argv vectors, stdout + stderr + exit codes byte-identical

held-out labels, authored blind

eval labels were written by reading source before the ranker ever ran on them — so the eval is allowed to say the ranker is wrong. It has.

sanitizer wall

ASan + UBSan + integer + float-cast, no-recover; 16 libFuzzer targets over its own readers of bytes it did not create; ThreadSanitizer nightly on main; leak suppression is one pinned file

its own output is audited

the showcase capture is regenerated and adversarially re-read as a claims corpus — the methodology ships in docs/METHODOLOGY.md

// the reason to believe any number on this deck

Claims you can trust, because we publish what failed

807

gate scripts named by test/regression.sh — and the COUNT is the counts snapshot 2026-11-13 of that loop, checked exact at every refresh and release

8

registered NEGATIVES — changes built, gated green, measured against a band written before the code, and reverted rather than tuned

0

numbers on this deck without a committed instrument behind them — §8 lists the ones this project refuses to publish at all

the fresh-clone arm	our own battery only ever ran inside a configured, long-lived worktree — one value of every AMBIENT input. It clones HEAD into a genuinely unconfigured checkout and re-runs the gates whose past failures were exactly that: a developer's git config, a fixture's inherited branch name, the checkout's own path length.
the merge-conservation gate	consistency is not conservation. A conflict resolution can drop a gate from the runner's loop while regenerating the pinned count from that SAME damaged list — both sides then agree at a wrong number. On a merge, the loop must equal the UNION of both parents', minus explicit tombstones.
the eight, by name	r7's prose-strip fix (predicted 12/22, band 10–14, measured 5/22) · RTA-lite, refuted twice · file-level evidence pooling, +0.00pp held-out · un-guarded query stemming · its IDF-guarded retry, VOIDED by its own verifier · query-term density weighting, both displacement guards tripped · --anchor and --cochange-boost, no confirmed lift · a name-coverage floor that MET its pre-registered band and was reverted anyway for a standing requirement.

A gate that cannot go red is a gate that proves nothing — which is why every arm on this deck ships with the mutation that makes it fail.

// own the losses — they are in the README too

Where it loses, published with the wins

The public C++ number is lower than the private one it replaced	SFML: strict file@10 28.7%, any@10 41.7%, first-hit MRR 0.21. Until 2026-08-07 this bullet compared against a ~89% any@10 private corpus that is no longer reproducible. That comparison is retired; the public number is the baseline going forward.
--grep costs more than it saves	+19.7% tokens / −11.2% — published next to the flag it indicts, and carrying the same private-corpus caveat as every figure in that source
PageRank is the wrong co-change ranker	3.8% recall@5 against 40.3% for plain lexical, and fusing the two made it worse — relatedness is lexical, importance is structural
Strict multi-file localization stays hard	single-file gold 73.4% vs multi-file 18.2% (held-out); the same cliff on every corpus — open headroom, said plainly

A tool that publishes its counterexamples is a tool whose wins you can believe.

// the single command for "is this code actually rotten?"

Six families of evidence, counted — never blended

structural	shape: complexity, size, nesting, params, readability rank	`--metrics`
lexical	identifier text: the naming rules	`--lint  --naming-consistency`
confusion	syntax: the atoms-of-confusion rules	`--lint`
historical	git change frequency, measured PER FILE	`--hotspots`
colocation	how much you must READ outside this file	`--context-ratio`
state	this function's OWN BODY touching non-local mutable state	`--nonlocal-state`

Ranked by the COUNT of distinct families that fire — never a weighted composite, because a composite lets one loud family impersonate six.

4 of 6 is what the strict preset counts — the verb says so itself, families="6" enabled_n="4". Two families did not clear the stability ladder.

`--quality-panel[=strict|default|lenient]` — and `--quality-delta` for the narrower question: what did MY change make worse?

// built for the agent's seat

One command wires it into your agent

```
$ ripwire wrap claude
```

claude · cursor · codex · windsurf · gemini · aider · opencode · openclaw · hermes — or --all to detect every one you have installed

33 MCP verbs

17 read verbs mirroring the CLI, 13 flagship reflexes (impact, uses, affected, edit_check, from_trace, connect …), 3 span-addressed edit verbs with a safety contract

lazy-body handles

read verbs return signatures and a stable handle; the agent fetches a body only when it decides it needs one — names by default, bytes on request

18 agent skills

moment-matched workflows (orient, navigate, change-check, quality-bar …) — wrap prints the recipe, skills/install.sh installs them

13 orchestrator loops

copy-paste prompts in prompts/: run the same audit, eval and head-to-head machinery that built this tool, on your own repository

wrap does not just print JSON: it emits the config in that agent's own shape with a use-when blurb per verb. Since 0.6.10, wrap claude routes edits the way Claude Code works: take the symbol's line range from ripwire, Read only that range, then the native Edit.

the MCP server exposes the same deterministic engine — one index, shared with the CLI, staleness-checked

// in the field: one engagement, the agent's own report, an older version

What it did for an orchestrated fleet of coding agents

`--callers`

one call redirected a task

A task was chartered to switch on a behaviour the team believed was live. One --callers call showed it had no production caller, and the task moved to the real gap: the data it needed had never been wired.

`--edit-check`

a silent divergence, caught early

A widely shared computation was refactored. --edit-check flagged incompatible call sites that a text search had missed; left alone, they would have drifted from the canonical path the day the feature was enabled.

`--quality-delta`

quality became a protocol

Run at each agent's "done", it made fix-or-justify the habit of agents that shared no context. Duplicated routines and near-identical twins were fixed, not waived.

`the honest boundary, from the same report`

The deepest findings did not come from the tool. They came from measurement the agents built themselves: worktree diffs, attribution sweeps, funnels. The tool was a floor for orientation and honesty, not a substitute for measurement design. And one known false-positive mode: --edit-check's arity heuristic over-counts where trailing parameters have defaults, and compiling settles it, as its own notes say.

source: README.md field report (#361) · one engagement · the agent's own report · an older version · testimony, not measurement

// it does not only read

Span-addressed edits: the map writes back

Three verbs, one resolved span

```
--replace-symbol-body=TARGET
--insert-before-symbol=TARGET
--insert-after-symbol=TARGET
  --edit-payload=FILE|-
```

TARGET is a name, an @FILE:LINE seed — paste the location out of a diff hunk or a compiler error and it binds the innermost enclosing definition — or a freshness-pinned handle from --grep --handles. No line numbers to keep in your head, and no whole-file rewrite.

The refusals are the feature

Freshness hash, lock, a re-check immediately before the rename, fsync, mode preserved, atomic rename. A target that resolves to more than one definition refuses. An empty payload refuses rather than being read as a deletion. Every refusal leaves the file byte-identical, and every success prints a JSON receipt whose span is the POST-EDIT byte range — where the payload now sits, not where it was aimed.

Preview the bytes before they exist

--edit-check=SYM --edit-payload=- --dry-run splices the payload over the span IN MEMORY, re-parses that one file, rebuilds the call graph over the re-derived tree, and answers the contract question about bytes nobody has written: did the signature change, and which callers does it break? Preview, then apply — no Read of the file in between.

Several edits, one transaction

--edit-plan=FILE is a versioned JSON multi-edit plan and its mode is EXPLICIT: exactly one of --dry-run or --apply, and neither is not a default. Every target, payload and span is preflighted before any write; overlapping edits refuse; payload paths are confined to the plan's own directory, so an absolute path or a '..' escape refuses and names what it resolved to.

Same engine on both seats: the three MCP edit verbs are these three flags, and the navigation the deck has no room for is the same story — --graph-query, --whereis, --stray-content, --at, --arch, --owners, --export, --help-task all sit in docs/COMMANDS.md with a recorded invocation and its output.

every claim on this slide is stated by `ripwire --help` and captured in docs/COMMANDS.md — test/deckcheck.sh proves each flag named here exists in the shipped binary

// standing on giants

The research inside — classic and current

49 repositories + 76 papers folded · and a labelled survey of 237 tools that contributed nothing, which says so — every row with the lesson taken and where it lives: docs/LINEAGE.md

1976	**Cyclomatic complexity — McCabe**	the metric behind --hotspots
1994	**Okapi BM25 — Robertson · Spärck Jones**	the lexical ranker (twice: subtoken+body, whole-name)
1998	**Personalized PageRank — Page · Brin**	the headline importance signal
2008	**Louvain modularity — Blondel et al.**	the module / community map
2009	**Reciprocal Rank Fusion — Cormack et al.**	deterministic signal fusion (--rank-by=rrf)
2011	**Readability model — Posnett · Hindle · Devanbu**	the lexical family of the quality panel
2019	**Delta Maintainability Model — di Biase et al.**	--dmm: one comparable number per change

TDAD · arXiv 2603.17973

a static test-map cut agent-caused regressions 6.08% → 1.82%; prose TDD instructions alone made agents WORSE → the shape of --test-gate

What-to-Retrieve · arXiv 2503.20589

retrieval selection beats retrieval volume for coding agents → the selection-over-dumping thesis

LocAgent / Loc-Bench (2025)

the localization metric (strict Acc@k) and the frozen 560-instance dataset every accuracy number here is scored on

scip-clang · Serena / clangd

the compiler-grade oracle round 9 graded both tools against — an outside referee, not a rival

tree-sitter

the incremental GLR parsing substrate — 27 grammars vendored, one parser per thread

counts gated in-repo: readmedriftcheck arm E derives them from LINEAGE.md's own tables — the lineage is the design rationale

// do not take any of it on trust

Every claim, and the command that re-derives it

194 long flags · 36 slides	`bash test/deckclaimcheck.sh`
every --flag named here exists	`bash test/deckcheck.sh`
74.7% fewer element bytes	`bash test/showcasecapturecheck.sh`
807 gate scripts (counts snapshot 2026-11-13)	`bash test/manifestcheck.sh`
49 repos · 76 papers · 237 surveyed	`bash test/readmedriftcheck.sh`
the ten moments, any row	`ripwire . --callers=SYM | wc -c`
the head-to-head table	`bench/headtohead/r4-2026-08-06/`
the oracle round	`bench/headtohead/r9-2026-08-09/RESULTS.md`
answers by release (in-sample)	`README.md, Measured — table and footnote`
memory by release (percent)	`docs/EVALS.md, Memory by release`

The deck is generated, and the generator is gated. present/deck5_ripwire_build.js is scanned for fabricated flags and its slide count is derived from its own addSlide() calls — a deck that drifts from the binary fails the suite.

docs/EVALS.md is the source of truth; §7 is the counterexamples, §8 the claims this project refuses to make

// sixty seconds to the first ranked map

Quickstart

```
git clone <repo> && cd ripwire
cmake -S . -B build && cmake --build build -j     # hermetic: works with the network off
./build/ripwire --help
```

`ripwire .`	the ranked map — start here on any unfamiliar repo
`ripwire . --for="cache invalidation"`	the task lens: what to touch, ranked
`ripwire . --callers=someFunction`	who calls it — with the honesty marker attached
`ripwire . --quality-panel`	is this code actually rotten? six families, one shortlist
`ripwire . --test-gate`	before you commit: exactly which tests must run
`ripwire wrap claude`	wire it into your agent, in that agent's own config shape

ripgrep for the speed. **tripwire** for the honesty. Apache-2.0.